# Mina Yeh

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## EDUCATION

**Cornell University, June 2029**, *Ithaca, NY*

- **3.93/4.00 | College of Engineering:** Electrical & Computer Engineering | Minor: Robotics
- **Awards & Honors:** Best Overall @ Cornell Makeathon 2026, Team Lead. Professional SFWA-Qualifying Fiction Publication @ OTHERSIDE.
- **Relevant Coursework:** Digital Logic, Signals & Systems, Circuits, Object-Oriented Programming, Differential Equations

**Stuyvesant High School, June 2025**, *Manhattan, NY*

- **GPA: 97.07/100** (Unweighted); Mu Alpha Theta, Rowing Captain; Parliamentary Debate Finalist; National Merit Finalist
- **Relevant Coursework:** Linear Algebra, Organic Chemistry, Existentialism
- **Awards & Honors:** Distinguished Honor Roll (Top 1%) on AMC (American Mathematics Competition)

## PROJECT EXPERIENCE

**Electrical Subteam** **February 2026 - Present**

*Cornell University Physical Intelligence*
*Ithaca, NY*

- Currently leading FPGA integration on Autonomous Hexapod computer vision.
- Developing an embedded C firmware stack using STM32 HAL/CMSIS, covering clock/GPIO initialization, FDCAN drivers, interrupt-driven communication, bounded TX/RX queues, message serialization, heartbeat monitoring, bus-off recovery, and USB-to-CAN forwarding.
- Created a PCB for USB-to-CAN transmission in Altium Designer, taking input from the Jetson t5000 and converting to three separate FDCAN buses for autonomous hexapod actuator control. Designed around the STM32G4 MCU.

**Single Cycle RISC-V Processor** **April 2026**

- Designed from scratch in **Verilog/SystemVerilog**, implementing the datapath and control logic. Verified with full written testbench in assembly and HDL; analyzed critical-path timing, synthesis results, and resource utilization in Quartus.

**VAMP - Digital Synthesizer** **July 2026**

- A digital sampler synthesizer with an on-board stereo microphone, a full octave of control, and instrument level, line level, and MIDI I/O ports. Built around the Daisy Seed, a STM32 based audio development board. Designed in KiCad. Firmware written with C++ and libDaisy.
- Fully documented in Github: https://github.com/syntonos/VAMP

**Blackjack as a Markov Decision Process** **June 2025**

- Blackjack, modeled as a Markov Decision Process, selecting for optimal play; live presentation on Linear Algebra optimization and MDP iteration, included discussion on real-world strategies for winning Blackjack. Coding demonstration in Python with numpy.

**Cornell FDSC 2300 - Culinary Science Website** **February 2026**

- Created and currently maintaining the Cornell University FDSC 2300 course website. React, TSX, frontend.

## ACTIVITIES

- **EasyEDA/JLCPCB,** Student Ambassador August 2026 - Present
- **Cornell Duffield Engineering,** Student Ambassador August 2026 - Present
- **qSTEM at Cornell,** Head of Outreach November 2025 - Present
- **Maker Lab, Cornell Taekwondo, Women in Computing at Cornell, Shakespeare Troupe, Rock Climbing Club**

## SKILLS & INTERESTS

- **Software & Tools:** LTSpice, Quartus, Altium, KiCAD, Fusion360, Onshape, CNC, Soldering
- **Languages:** SystemVerilog, Verilog, C++, Python (NumPy, Matplotlib), Java, Javascript (React), MATLAB
- **Foreign Languages:** Mandarin (Professional Proficiency)